UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 004 · Implicit functions, problems on implicit and composite functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9 · S003·26	Deck 004 / 003, slide number (rendered .pptx)
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
T003b 00:10:33	Transcript 003 part II (implicit problems at its end)
CW U1 P17 · ESA Jan-24 s3 1d	Unit 1 class-work problem 17 · past-paper question

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A Formula sheet

1 Implicit functions, level surfaces and the differential

E1.1 · Total differential ESA ★★

$$dz = f_x(x,y)\,dx + f_y(x,y)\,dy$$

GIVES Approximate change in z for small changes dx, dy.

APPLIES $z = f(x,y)$ differentiable.

NOTE n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

2 Implicit differentiation in two variables

E2.1 · Implicit differentiation, two variables ESA ★★

$$f(x,y) = c \implies \frac{dy}{dx} = -\frac{f_x}{f_y}, \qquad f_y \neq 0$$

GIVES Slope of the level curve at a point.

APPLIES $f_y \neq 0$ at the point.

NOTE From $df = f_x\,dx + f_y\,dy = 0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E2.2 · Tangent line at a point ESA ★★

$$y - y_0 = m\,(x - x_0), \qquad m = \left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

GIVES Equation of the tangent to the curve at (x_0, y_0).

APPLIES (x_0, y_0) on the curve.

NOTE Through the origin: $y = mx$.

S004·7 · T004a 00:14:25

E2.3 · Derivative of a constant power and a variable power ESA ★★

$$\frac{d}{dx}a^x = a^x \ln a, \qquad \frac{d}{dx}x^n = n\,x^{n-1}$$

GIVES The two rules needed for $(\cos x)^y$ and $(\sin y)^x$.

APPLIES In a partial derivative, the other variable is a constant.

NOTE $\ln$ needs a positive argument: $\cos x > 0$, $\sin y > 0$.

S004·8 · T004a 00:16:43

3 Implicit functions of three variables

E3.1 · Implicit differentiation, three variables

ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

GIVES Partial derivatives of the implicitly defined variable.

APPLIES Hold y constant for z_x; hold z constant for y_x.

NOTE Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

4 Composite functions and the chain rule

E4.1 · Chain rule, two intermediate variables

ESA ★★★

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

GIVES Partial derivatives of a composite function.

APPLIES $z=f(x,y)$, $x=g(u,v)$, $y=h(u,v)$.

NOTE All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E4.2 · Chain rule, three intermediate variables

ESA ★★★

$$\frac{\partial v}{\partial x}=\frac{\partial v}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial v}{\partial s}\frac{\partial s}{\partial x}+\frac{\partial v}{\partial t}\frac{\partial t}{\partial x}$$

GIVES One term per intermediate variable.

APPLIES $v=f(r,s,t)$, each of r,s,t a function of x,y,z.

NOTE A term is 0 when that intermediate variable does not contain x.

S003·15 · T003a 00:23:10

E4.3 · Function of one intermediate variable

ESA ★★

$$u=f(r),\ r=\sqrt{x^2+y^2+z^2} \implies \frac{\partial u}{\partial x}=f'(r)\,\frac{\partial r}{\partial x}=f'(r)\,\frac{x}{r}$$

GIVES $\partial u/\partial x$ in terms of $f'(r)$.

APPLIES u depends on x,y,z only through r.

S004·12 · T004b 00:05:07

5 Composite functions in polar coordinates

E5.1 · Polar coordinates: derivatives you reuse

ESA ★★

$$\begin{aligned} &x = r\cos\theta,\ y = r\sin\theta: && x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta \\ &r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: && r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2} \end{aligned}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r,\theta)$.

S004·13 · S004·15 · S004·17

E5.2 · Polar identity

ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

NOTE Proved in W5.6.

CW U1 P15

B Symbol glossary

Symbol	Name	Meaning	Appears in
$f(x,y)=c$	Level curve / implicit relation	Defines y as a function of x	E2.1
$f(x,y,z)=0$	Level surface	Defines z as a function of x, y	E3.1
$c,\ c_1$	Level	The constant value of f on one level surface	E2.1
f_x, f_y, f_z	Partial derivatives of the relation	Numerator and denominator of the implicit formulas	E2.1, E3.1
df, dx, dy	Differentials	$df = f_x\,dx + f_y\,dy$; zero on a level curve	E1.1
$\dfrac{dy}{dx}$	Slope of a level curve	$-f_x/f_y$	E2.1, E2.2
m, (x_0, y_0)	Slope, point of contact	Tangent $y - y_0 = m(x - x_0)$	E2.2
$u = f(x,y)$	Composite function	Outer function of the chain (unknown f kept as f_x, f_y)	E4.1
s, t / u, v	New variables	Final variables of the composite function	E4.1
p, q, w	Named arguments	e.g. $p = x - y$, $q = y - z$, $w = z - x$	E4.2
r, θ	Polar coordinates	$x = r\cos\theta$, $y = r\sin\theta$	E5.1, E5.2
r (slide 004·12)	Distance from the origin	$\sqrt{x^2+y^2+z^2}$	E4.3
$f'(r)$	Ordinary derivative	u depends on x, y, z only through r	E4.3
$\ln$, $\log$	Natural logarithm	Appears from a^x and from x^x	E2.3

Remark. The letter f means two different things in this class: the implicit relation $f(x,y)=0$ (topics 1–3) and the outer function of a composite $u = f(x,y)$ (topics 4–5). Likewise r is a radius in polar problems but $r = \sqrt{x^2+y^2+z^2}$ in slide 004·12.

C Which formula, which symbol?

Relation given	You want	Formula
$f(x,y)=c$	$\dfrac{dy}{dx}$	$-\dfrac{f_x}{f_y}$
$f(x,y,z)=0$	$\dfrac{\partial z}{\partial x},\ \dfrac{\partial z}{\partial y}$	$-\dfrac{f_x}{f_z},\ -\dfrac{f_y}{f_z}$
$f(x,y,z)=0$	$\dfrac{\partial y}{\partial x}$ (z held)	$-\dfrac{f_x}{f_y}$
$u=f(x,y)$, x,y of s,t	$\dfrac{\partial u}{\partial s}$	$f_x\,x_s+f_y\,y_s$
$u=f(x,y)$, with x,y linked by $g(x,y)=0$	$\dfrac{du}{dx}$	$u_x+u_y\dfrac{dy}{dx}$ with $\dfrac{dy}{dx}=-\dfrac{g_x}{g_y}$